[FULL NAME]

BACKEND / ML / AI-OPERATION INTERN

[Phone Number] | [Email Address]
Vietnam | [GitHub / Portfolio Profile]

CAREER OBJECTIVE

Final-year Computer Science student at PTIT seeking a backend, ML/NLP or AI-operation internship where I can apply Python-based AI pipelines and Java application development in a production environment. I bring hands-on experience building and comparing deep learning models (TrOCR, CRNN), OCR-based NER systems for invoice processing, and full-stack desktop tools — and I am eager to grow alongside an engineering team on real-world projects.

EDUCATION

Posts and Telecommunications Institute of Technology (PTIT) — Computer Science 2022 – 2027

• **Recent GPA:** 3.18 / 4.0 • Academic excellence scholarship recipient.

TECHNICAL SKILLS

Languages: Python, Java, JavaScript, SQL

ML / AI: PyTorch, HuggingFace Transformers, ONNX Runtime, TensorFlow (training), PaddleOCR / PaddlePaddle, OpenCV

Web & Backend: Streamlit, Flask, Node.js / Express, HTML5 / CSS / JS

Java Ecosystem & Tools: Swing, JNA, JUnit 5, Maven, FlatLaf, Gson | Git, Linux, Jupyter Notebook, Ollama

AI & Productivity Tools: Proficient with Claude, Gemini, and ChatGPT for research, code assistance, and workflows | MS Office

Foreign Language: English — **IELTS 7.0** (Listening 8.0 · Reading 7.5 · Writing 6.0 · Speaking 6.0)

PROJECTS

Handwriting Recognition 04/2026 – Present

Python · PyTorch · HuggingFace Transformers · ONNX Runtime · OpenCV · Streamlit | [Repository Link Available]

- Dual-backend recognition app comparing TrOCR (334M params, 1–3 s/line) and a custom-trained CRNN exported to ONNX (1.5 MB, ~20 ms/word).
- Trained CRNN on the IAM Handwriting Dataset; achieved ~12% CER after 10 epochs, ~7% CER after 40–50 epochs.
- Built Streamlit web UI with live drawable canvas, URL/clipboard paste, multi-line paragraph segmentation, and confidence scoring.
- Engineered robust preprocessing pipeline: deskew, CLAHE contrast enhancement, bilateral denoise, perspective correction, adaptive binarization.

Invoice NER — Masking System 04/2026 – Present

Python · PaddleOCR · PaddlePaddle · Streamlit · NumPy · Pillow | [Repository Link Available]

- NLP project: end-to-end pipeline to detect and label named entities (fields) on invoice images using PaddleOCR 3.0 for text extraction and a custom NER tagging layer.
- Multi-stage processing pipeline: OCR → token segmentation → entity classification → structured output — with results published to a Streamlit web UI for verification and correction.
- Adopted Cookiecutter-style project layout (data, src, notebooks, models, reports, tests) for full reproducibility.

Keyboard Remapper 04/2026 – 05/2026

Java 11 · JNA · FlatLaf · Gson · Ollama · Node.js / Express · HTML5 Canvas | [Repository Link Available]

- Full-featured Windows keyboard remapper using low-level OS hooks (JNA WH_KEYBOARD_LL) for system-wide key interception.
- Macro recording/playback with configurable repeat modes, profile import/export (.bkm), and AI assistant (KeySup) powered by local Ollama (Gemma3) with regex fallback.
- Browser-based remote control UI built with HTML5 Canvas; dual web server (Python stdlib + Node.js/Express) bridging to the Java app over HTTP.
- Security hardening: CSRF protection, XSS sanitisation, Content Security Policy, atomic file writes. JUnit 5 unit tests included.

ACTIVITIES & LEADERSHIP

Secretary | Youth Union 2022 – 2023

- Managed records, task assignment, and monthly/quarterly/annual activity planning.
- Political education: disseminating policies and guidance to union members.
- Identified and mentored outstanding union members for development.

SOFT SKILLS & WORKING STYLE

- **Communication:** Clear written/verbal communication across technical & non-technical audiences.
- **Teamwork & Collaboration:** Experienced leading small groups and contributing as an individual.
- **Time Management:** Plans work by sprint; consistent and methodical with high reliability.
- **Self-motivated:** Proactively tackles new topics and tools without waiting to be directed.